

Mapping the Design Space of Task-Priority Resource Allocation in Multi-Task Wireless Semantic Communication

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Abstract—Multi-task semantic communication systems must allocate a finite channel resource across tasks that differ in operator importance and channel-state sensitivity. Existing multi-task designs either share the decoded representation symmetrically, which forces the system to operate at the worst-task rate, or hard-partition by task identity, which forgoes cross-task sharing. Neither approach acknowledges that tasks have unequal priority and unequal noise sensitivity. We introduce Task-Priority Semantic Resource Allocation (TP-SRA), a framework that formalizes priority-aware allocation as a constrained optimization with a water-filling-type characterization of the optimum. The proposed solver is a small SNR-conditioned learnable gate, trained jointly with the encoder and decoder, that allocates the shared decoded representation across tasks based on operator priorities and the current channel state. We then map the practical design space by comparing soft multiplicative masks against per-task Houshy adapters at matched parameter budget, and against a runtime-conditional variant trained under stochastic priorities. Each design is evaluated across AWGN, Rayleigh, and Rician channels under paired-bootstrap confidence intervals to identify where priority-aware allocation yields reliable cross-channel gains and where it does not. Our findings show that a simple hard-partition baseline is the only mechanism delivering consistent cross-channel improvements under multi-seed evaluation, that runtime priority interfaces collapse into multi-task regularizers rather than steerable controls, and that feature-mixing readouts outperform feature-scaling gates on the hardest task. A controlled rate-scaling experiment further shows that the value of the allocation mechanism is rate-dependent: increasing the channel budget without controlling allocation hurts performance, while the gated variant absorbs the rate increase and widens its gap to the un-gated baseline.

Index Terms—Semantic communication, multi-task learning, resource allocation, financial NLP, FinBERT, DeepSC.

I. INTRODUCTION

Semantic communication has emerged as a paradigm in which the transmitter encodes the meaning of a source signal rather than its raw bits, with the goal of reducing channel-symbol cost while preserving downstream task performance [1]. Architectures such as DeepSC [2] and its multi-task extensions [3] have demonstrated that joint training of source coding, channel coding, and task heads yields large gains over modular pipelines, particularly at low SNR.

An open question in multi-task semantic communication is how the shared decoded representation should be allocated across tasks. Existing multi-task designs [3], [8] either share it symmetrically or hard-partition by task identity, overlooking two operational realities: tasks have unequal operator importance and unequal noise sensitivity. Symmetric allocation forces operation at the worst-task rate, while hard partitioning forgoes cross-task sharing. We address this gap by formulating allocation as a constrained optimization with a closed-form characterization, realized by a family of allocation interfaces of which a channel-state-aware gate is one instantiation, trained jointly with the encoder and decoder.

TP-SRA is distinct from joint source-channel coding (JSCC), which optimizes a single-task end-to-end signal pipeline without operator priorities or per-task allocation, and from rate-distortion-optimal coding, which characterizes achievable distortion at fixed rate for a single source. TP-SRA instead distributes a finite semantic resource across heterogeneous tasks under a priority-weighted objective with data-dependent per-task utilities. Theorem 2 extends this picture to a joint physical and semantic allocation analogous to multi-carrier power-rate water-filling.

A. Contributions

We introduce Task-Priority Semantic Resource Allocation (TP-SRA), a framework for allocating a finite channel budget across tasks under operator-specified priorities and channel-state awareness. We show when priority-aware allocation yields reliable gains, when it does not, and connect these empirical findings to theoretical limits on optimal allocation. Our contributions are the following.

- 1) Empirical finding on hard partitioning. Across AWGN, Rayleigh, and Rician channels with three random seeds per method, only B1 (hard equal split) achieves a bootstrap-significant ESG gain on every channel; no soft-mask variant (B2 or B3) reaches significance on any channel, and the joint rate-mask solver (B6) does not close the cross-channel gap.
- 2) Two-axis water-filling characterization. Theorem 1 characterizes the mask-only optimum, while Theorem 2

extends it to the joint rate-mask problem with the channel rate $\|\mathbf{c}\|_1$ as a free variable, yielding two coupled marginal-equalization conditions, one over tasks (semantic axis) and one over symbols (physical axis). Methods that hold $\mathbf{c} \equiv \mathbf{1}$ optimize only the semantic axis, which explains why fixed-rate gates do not robustly transfer across fading.

- 3) Design space taxonomy. A systematic comparison of soft masks (B3), per-task Housby adapters (B4), and the runtime priority interface (B3-cond) under paired-bootstrap confidence intervals (Table IV) shows that adapters beat the gate on the hardest task at matched parameter budget, that Dirichlet stochastic-priority training collapses the runtime priority lever to a sub-three percentage-point range, and that training-time priority specialization yields a directional Pareto trade-off whose magnitude on the prioritized task is variance-dominated at four seeds, correcting an earlier single-seed report.

B. Paper Organization

Section II formalizes the system model and performance assumptions. Section III presents TP-SRA, including Theorems 1 and 2 and the SoftMaskGate solver. Section IV describes the FinBERT-DeepSC testbed, channels, and training protocol. Section V reports cross-channel, priority-lever, design-space, and joint rate-mask results. Section VII concludes.

II. SYSTEM MODEL

The pipeline is shown in Fig. 1. The bottleneck is the channel vector, while the masks of §III act on the decoded representation. The framework is task-agnostic; the financial-NLP instantiation is described in Section IV.

A. Abstract Multi-Task Semantic Channel

Let $x \in \mathcal{X}$ be a source sample and $\mathcal{T} = \{1, \dots, T\}$ index the tasks. A semantic encoder $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ ($d=768$) produces $\mathbf{z} = f_\theta(x)$; a channel encoder $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^N$ projects to $\mathbf{s} = g_\phi(\mathbf{z})$ under power normalization $\mathbb{E}\|\mathbf{s}\|_2^2 = N$. The channel produces $\hat{\mathbf{s}} = \mathbf{h} \odot \mathbf{s} + \mathbf{n}$ with $n_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, \sigma^2)$ and $\text{SNR}_{\text{dB}} = -10 \log_{10} \sigma^2$. We instantiate AWGN ($h_i \equiv 1$), Rayleigh ($h_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, 1)$), and Rician ($K=4$). A shared channel decoder $g'_{\phi'} : \mathbb{R}^N \rightarrow \mathbb{R}^d$ feeds T task-specific heads $\{h_{\psi_t}^{(t)}\}$ producing predictions $\hat{y}^{(t)}$. The physical channel budget N is fixed by deployment; this paper addresses how the shared d -dimensional decoded representation is allocated across the T downstream tasks.

B. Resource Allocation as a Mask Operation

We model the allocation as a per-task soft mask $\mathbf{m}^{(t)} \in [0, M]^d$ ($M \geq 1$) applied after the channel decoder,

$$\mathbf{z}^{(t)} = \mathbf{m}^{(t)} \odot g'_{\phi'}(\hat{\mathbf{s}}), \quad (1)$$

with $\hat{y}^{(t)} = h_{\psi_t}^{(t)}(\mathbf{z}^{(t)})$. The optimization variables are the masks $\{\mathbf{m}^{(t)}\}$, not the channel budget N . The per-task L1

share defines $N_t \triangleq \|\mathbf{m}^{(t)}\|_1 \cdot N / \sum_k \|\mathbf{m}^{(k)}\|_1$ with $\sum_t N_t = N$ as a normalized accounting unit for the analysis of Section III. TP-SRA therefore operates at the semantic resource layer (representation dimensions across tasks), complementary to physical-layer water-filling on sub-channel power.

C. Performance Model

Let $\mathcal{L}_t(N_t, \sigma^2)$ denote the expected per-task loss. We assume: (A1) $\mathcal{L}_t(\cdot, \sigma^2)$ non-increasing in N_t ; (A2) $\mathcal{L}_t(N_t, \cdot)$ non-decreasing in σ^2 ; (A3) $\mathcal{L}_t(\cdot, \sigma^2)$ convex in N_t ; (A4) $\mathcal{L}_t$ jointly smooth and convex in $(\mathbf{m}^{(t)}, \mathbf{c})$ (used only by Theorem 2). Conditions (A1) and (A2) are standard monotonicity, while (A3) follows from a saturation argument: the rate- N_t task-relevant information is upper-bounded, so additional dimensions yield diminishing returns. Single-task sweeps over $N \in \{4, 8, 16, 32\}$ confirm the diminishing-returns shape. The operator assigns priorities $\mathbf{w} = (w_1, \dots, w_T)$ with $w_t \geq 0$ and $\sum_t w_t = 1$.

III. TASK-PRIORITY SEMANTIC RESOURCE ALLOCATION

Building on the mask-based allocation surrogate of §II-B and the performance assumptions of §II-C, we now formulate Task-Priority Semantic Resource Allocation (TP-SRA): we cast the mask-design problem as a constrained optimization, characterize its optimum in closed form under an idealized model, and describe an SNR-conditioned learnable gate that solves the mask problem end-to-end.

A. The Allocation Problem

Working in the scalar-allocation surrogate of §II-B (where N_t is the L1 share of the decoded representation allocated to task t), the TP-SRA allocation problem under channel state σ^2 and priorities $\mathbf{w}$ is

$$\begin{aligned} \min_{\{N_t\}_{t=1}^T} \quad & \sum_{t=1}^T w_t \mathcal{L}_t(N_t, \sigma^2) \\ \text{s.t.} \quad & \sum_{t=1}^T N_t = N, \quad N_t \geq 0. \end{aligned} \quad (2)$$

The implemented system optimizes directly over the masks $\{\mathbf{m}^{(t)}\}$; (2) serves as the characterization that motivates the gate's design (§III-D).

B. Closed-Form Optimal Allocation (Idealized Model)

We first analyze an idealized model in which tasks compete for a shared scalar budget under the loss curves $\{\mathcal{L}_t\}$. This model yields a closed-form priority-weighted marginal-equalization rule, that is, a rule that equalizes the priority-weighted marginal return of each additional symbol across tasks (Theorem 1), and that motivates the priority-conditional form of our learned gate. The theorem characterizes the optimal policy under assumptions (A1)–(A3); the implemented SoftMaskGate (§III-D) does not solve (2) literally, but realizes the same design intuition by learning an end-to-end approximation of the optimal mapping.

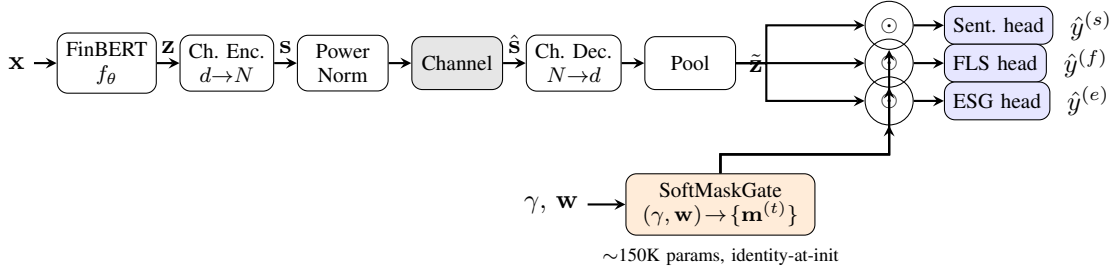


Fig. 1. TP-SRA architecture (FinBERT-DeepSC, $d=768$, $N=32$). The pooled decoded representation $\tilde{\mathbf{z}}$ is masked per-task by the SoftMaskGate, conditioned on the channel SNR estimate γ and operator priorities $\mathbf{w}$. The gate is the only TP-SRA-specific component.

Theorem 1 (Generalized TP-SRA Allocation). *Under assumptions (A1)–(A3), the unique optimal solution $\{N_t^*\}_{t=1}^T$ to (2) satisfies the water-filling-type condition*

$$-w_t \left. \frac{\partial \mathcal{L}_t}{\partial N_t} \right|_{N_t^*} = \lambda^*, \quad \forall t \in \mathcal{T}, \quad (3)$$

where $\lambda^* \geq 0$ is the Lagrange multiplier of the budget constraint, chosen so that $\sum_{t=1}^T N_t^* = N$. The result follows from applying KKT to the convex sum over a convex polytope, with complementary slackness $\mu_t = 0$ for $N_t^* > 0$.

The intuition is that optimal priority-aware allocation resembles classical water-filling: capacity flows preferentially to tasks with higher priority-weighted marginal utility, balanced so that all active tasks share the same marginal rate λ^* .

Remark 1. The left-hand side of (3) is a priority-weighted marginal task utility that plays the role of the channel-capacity gradient in classical water-filling. The analogy is structural; the equivalence to classical water-filling does not hold because $\partial^2 \mathcal{L}_t / \partial N_t^2$ is data-dependent rather than logarithmic. We therefore use Theorem 1 as a characterization and let a neural network solve the allocation directly from data (§III-D).

C. Joint Rate-Mask Allocation

The mask allocation in (2) holds the channel rate N fixed, treating it as an exogenous parameter set by the system designer. A natural extension is to make the channel rate itself a free variable, jointly optimized with the masks. Let $\mathbf{c} \in [0, 1]^{N_{\max}}$ be a soft per-symbol activation gate and let $\mathbf{s} = \mathbf{c} \odot g_\phi(\mathbf{z})$ be the post-gating transmitted vector. Its effective rate is $\|\mathbf{c}\|_1 \leq N_{\max}$, and the L1 term $\lambda_c \|\mathbf{c}\|_1$ in the objective prices each active symbol so that the optimizer trades rate against task accuracy. The joint problem is

$$\min_{\{\mathbf{m}^{(t)}\}, \mathbf{c}} \sum_{t=1}^T w_t \mathcal{L}_t(\mathbf{m}^{(t)}, \mathbf{c}, \sigma^2) + \lambda_c \|\mathbf{c}\|_1 \quad (4)$$

subject to the per-task mask constraints of (2), $\mathbf{c} \in [0, 1]^{N_{\max}}$, and the post-gating power constraint $\mathbb{E}\|\mathbf{c} \odot g_\phi(\mathbf{z})\|_2^2 \leq P_{\max}$.

Theorem 2 (Joint Rate-Mask Allocation). *Under (A1)–(A4), any optimum $(\{\mathbf{m}^{(t)*}\}, \mathbf{c}^*)$ of (4) satisfies two coupled marginal-equalization conditions,*

$$\begin{aligned} -w_t \left. \frac{\partial \mathcal{L}_t}{\partial N_t} \right|_* &= \lambda_N^*, \quad \forall t \in \mathcal{T}_+, & (\text{semantic axis}) \\ -\sum_{t=1}^T w_t \left. \frac{\partial \mathcal{L}_t}{\partial c_i} \right|_* &= \lambda_c, \quad \forall i : 0 < c_i^* < 1, & (\text{physical axis}) \end{aligned}$$

where $\mathcal{T}_+ = \{t : N_t^* > 0\}$ and $\lambda_N^* \geq 0$ is the Lagrange multiplier of the mask-budget constraint of (2). The semantic axis indexes per-task allocations on the learned representation, while the physical axis indexes per-symbol channel-rate gates.

The intuition is that optimality requires jointly equalizing priority-weighted marginal utilities across both per-task masks (semantic axis) and per-symbol channel-rate gates (physical axis), a coupling that fixed-rate single-axis methods cannot exploit.

Remark 2. Theorem 2 extends the marginal-equalization characterization across two axes, tasks (semantic) and symbols (physical), structurally analogous to joint power-rate water-filling in multi-carrier systems [9]. The methods evaluated in §V (Baseline, B1, B2, B3) hold $\mathbf{c} \equiv \mathbf{1}$, optimizing only along the semantic axis. The physical-axis condition (physical axis) is therefore generically violated at their solutions when $\lambda_c > 0$. This predicts the empirical pattern observed in §V-A: a gate with no physical-axis degree of freedom can only respond to channel state by reweighting the semantic axis, and an AWGN-tuned reweighting does not consistently differentiate from the un-gated baseline once fading breaks per-symbol equivalence. A learnable solver for (4) is instantiated in §V-D; the simplest architecture does not close the gap.

D. Learnable Gate (SoftMaskGate)

We propose a small SNR-conditioned neural solver that learns the masks $\{\mathbf{m}^{(t)}\}$ end-to-end. The gate is conditioned on γ because the reliability profile of decoded features is itself SNR-dependent: at low SNR, channel noise corrupts feature dimensions non-uniformly, so the optimal task-conditional readout must adapt to the channel state. Given an SNR

estimate $\gamma = -10 \log_{10} \sigma^2$, a two-layer MLP produces a residual $\delta(\gamma) \in \mathbb{R}^{T \times d}$ and the masks are formed as

$$\mathbf{m}^{(t)}(\gamma) = \text{clip}_{[0,M]}(\mathbf{1} + \delta_t(\gamma)). \quad (5)$$

The amplitude bound M (set to 2 in all experiments) prevents signal blow-up while allowing the gate to attenuate or amplify features by up to a factor of two. The clipping is non-differentiable at the boundary but acts as a soft constraint during training. The output layer is initialized to zero, so $\delta(\gamma) \equiv \mathbf{0}$ and $\mathbf{m}^{(t)}(\gamma) \equiv \mathbf{1}$ at the start of training. This identity-at-initialization property is critical: by zero-initialization the gated model is forward-equivalent to the ungated baseline at training step zero, which guarantees that fine-tuning from any pre-trained multi-task baseline cannot harm initial performance.

The gate adds approximately 150K parameters (less than 0.2% of FinBERT’s 110M) and negligible inference overhead, since the dominant cost is the FinBERT forward pass; gains therefore cannot be attributed to increased model capacity alone. Empirically, under uniform priorities the gate converges to a parsimonious near-shared SNR-modulator (mean pairwise mask cosine similarity 0.989), and per-task differentiation engages only when the priority weights $\mathbf{w}$ break symmetry, consistent with the marginal-equalization condition of Theorem 1. We train the gate once per priority vector $\mathbf{w}$ and report the resulting Pareto trade-offs in §V-B. We further consider a runtime variant (B3-cond) in which the gate accepts $(\gamma, \mathbf{w})$ as input and is trained under stochastic priority sampling $\mathbf{w} \sim \text{Dir}(\alpha)$ per batch, whose empirical behavior is reported in §V-B.

E. Method Family

We instantiate four design points along a progression from baseline to full gate. B0 is the symmetric-sharing baseline of [3], with $\mathbf{m}^{(t)} \equiv \mathbf{1}$. B1 (Hard Equal Split) uses disjoint d/T indicator masks. B2 (Soft Static) uses trainable but SNR-independent masks. B3 (Soft Dynamic) is the full SoftMaskGate with γ -conditioned masks. A B1 win indicates that the optimum lies near the hard-indicator corner of the mask polytope.

IV. EXPERIMENTAL SETUP

A. Testbed: FinBERT-DeepSC Multi-Task System

The testbed comprises $T=3$ tasks of differing class structure and difficulty: Sentiment (3 classes: positive/negative/neutral), FLS (forward-looking statement detection, 3 classes), and ESG (environmental, social, governance, none, 4 classes). The semantic encoder f_θ is FinBERT [4], a financial-domain BERT producing $d=768$ -dimensional sentence embeddings; the channel encoder/decoder pair follows DeepSC [2] with channel budget $N=32$; each task head is a single linear layer. The B0 baseline implements the symmetric-sharing multi-task SemCom paradigm of Sheng et al. [3]; B0–B3 share the same data, channel, optimizer, and SNR-sweep protocols, so performance differences are attributable to allocation rather than to backbone or training. The corpus is the Financial

PhraseBank [5] (4,837 expert-annotated sentences); ESG and FLS labels come from FinBERT-tone and FinBERT-FLS [6, 7], which induces a label-noise floor on those two tasks (Sentiment is unaffected).

B. Channel Models, SNR Sweep, and Training

We instantiate the three channels of §II (AWGN, Rayleigh, Rician with $K=4$). At evaluation we sweep $\text{SNR} \in \{-15, -10, -5, -2, 0, 2, 5, 10, 15, 20\}$ dB. Each of the 4×3 method-channel pairs is trained for 12 epochs with AdamW ($\eta=2 \times 10^{-5}$, batch 16, weighted cross-entropy to mitigate ESG class imbalance); training SNR is sampled uniformly from $[-10, 10]$ dB to expose the gate to the full operating range.

V. RESULTS AND DISCUSSION

A. Cross-Channel Method Comparison

Table I summarizes mean accuracy across the SNR sweep, fully multi-seeded ($n = 3$). Table II reports paired-bootstrap 95% confidence intervals on (method – Baseline) for ESG, the highest-variance and highest-imbalance task, on each channel. Resampling is performed over (seed, SNR) pairs jointly, so the reported CIs incorporate both training-seed variance and eval-sample variance. The seed count ($n=3-4$) reflects training cost, and paired-bootstrap CIs provide conservative uncertainty quantification.

B1 (hard equal split) is the only method whose ESG gain CI excludes zero on all three channels (+4.42 AWGN, +3.19 Rayleigh, +7.06 Rician, all 95% CIs strictly positive); B1 also stabilizes the AWGN standard deviation by a factor of four (1.51 vs. 6.55). The soft-mask gates (B2 and B3) tell the opposite story: under multi-seed evaluation, none of their ESG deltas reach significance on any channel (Table II). The runtime SNR-conditioned gate adds enough variance under multi-seed evaluation that any AWGN-specific specialization does not yield a consistent cross-channel benefit, a non-trivial observation that any priority-aware design has to reckon with.

Per Theorem 2, B0–B3 hold the channel rate fixed at $N = 32$ and optimize only the semantic axis; §V-F examines what happens when the channel rate is doubled, and §V-D tests the joint solver.

TABLE I
CROSS-CHANNEL FINBERT-DEEPCSC MEAN ACCURACY ACROSS SNR $\in [-15, 20]$ dB, IN %. ALL ENTRIES ARE MULTI-SEED ($n = 3$, MEAN $\pm$ STD). SEE TABLE II FOR PAIRED-BOOTSTRAP CIs ON THE ESG DELTAS.

Channel	Task	Baseline	B1	B2	B3
AWGN	Sentiment	96.73 $\pm$ 0.92	96.95 $\pm$ 0.83	97.09 $\pm$ 1.30	97.39 $\pm$ 0.86
	FLS	93.22 $\pm$ 2.20	91.68 $\pm$ 4.36	91.61 $\pm$ 1.55	92.46 $\pm$ 1.76
	ESG	69.94 $\pm$ 6.55	74.36 $\pm$ 1.51	72.06 $\pm$ 5.14	71.41 $\pm$ 3.23
Rayleigh	Sentiment	94.92 $\pm$ 0.60	94.39 $\pm$ 0.90	94.66 $\pm$ 0.89	95.09 $\pm$ 0.65
	FLS	91.84 $\pm$ 3.58	92.13 $\pm$ 2.95	90.00 $\pm$ 1.70	90.72 $\pm$ 3.92
	ESG	69.44 $\pm$ 7.57	72.63 $\pm$ 0.81	70.55 $\pm$ 4.29	69.08 $\pm$ 3.87
Rician	Sentiment	79.88 $\pm$ 0.93	80.49 $\pm$ 1.69	80.10 $\pm$ 1.65	80.78 $\pm$ 1.31
	FLS	73.61 $\pm$ 13.48	73.78 $\pm$ 10.93	74.18 $\pm$ 13.82	75.12 $\pm$ 11.23
	ESG	48.67 $\pm$ 0.59	55.73 $\pm$ 2.71	48.69 $\pm$ 1.51	47.20 $\pm$ 3.10

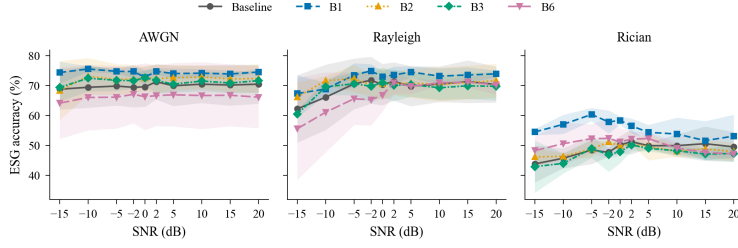


Fig. 2. ESG accuracy vs. SNR per method, per channel ($n=3$ seeds; shaded bands are ± 1 std). B1 dominates at low SNR on every channel and is strongest on Rician. Soft-mask gates (B2, B3) do not consistently improve over Baseline on fading channels. B6 tracks the soft-mask cluster rather than closing the gap.

TABLE II

PAIRED-BOOTSTRAP 95% CIs ON (METHOD – BASELINE) FOR ESG, PER CHANNEL, IN PP. * MARKS INTERVALS THAT EXCLUDE 0. B1 IS THE ONLY METHOD WHOSE ESG GAIN CI EXCLUDES ZERO ON ALL THREE CHANNELS. THE B6 ROW REPORTS THE JOINT RATE-MASK RESULT (§V-D); NEGATIVE VALUES INDICATE THAT B6 UNDERPERFORMS BASELINE.

Method	AWGN	Rayleigh	Rician
B1 Equal	+4.42* [+2.52, +6.46]	+3.19* [+0.70, +5.74]	+7.06* [+5.38, +8.87]
B2 Static	+2.12 [-0.77, +5.14]	+1.12 [-1.34, +3.72]	+0.03 [-1.62, +1.64]
B3 Dynamic	+1.47 [-0.94, +4.00]	-0.36 [-2.57, +1.99]	-1.46 [-3.22, +0.29]
B6 Joint	-3.67* [-5.02, -2.31]	-2.71 [-6.98, +1.16]	+1.60 [-0.39, +3.64]

The B1 advantage concentrates at low SNR (Fig. 2). On AWGN, restricting the mean to $[-15, -5]$ dB widens B1’s ESG gain over Baseline from +4.42 to +5.56 pp, while at > 10 dB methods converge. On Rician, B1’s lead persists across nearly the entire range. The soft-dynamic gate (B3) shows the same low-SNR emphasis on AWGN but inverts on fading.

B. Priority-Weight Sensitivity

All preceding results use uniform $w_t = 1/T$. To test whether the learned B3 gate responds to non-uniform priorities as Theorem 1 predicts, we retrain B3 on AWGN under $\mathbf{w}_{\text{ESG}} = (0.1, 0.1, 0.8)$ and $\mathbf{w}_{\text{SENT}} = (0.8, 0.1, 0.1)$ (Table III).

TABLE III

EFFECT OF PRIORITY VECTOR $\mathbf{w} = (w_S, w_F, w_E)$ ON B3 (AWGN, MEAN $\pm$ STD ACROSS SNR SWEEP, $n = 4$ SEEDS, IN %). PRIORITY-LEVER DELTAS ARE PAIRED-BOOTSTRAP 95% CIs (10,000 RESAMPLES); † MARKS DELTAS WHOSE CI EXCLUDES 0.

$\mathbf{w}$	Sent.	FLS	ESG
(1, 1, 1)	97.36 $\pm$ 0.42	90.30 $\pm$ 1.48	68.56 $\pm$ 4.61
(0.1, 0.1, 0.8)	94.51 $\pm$ 1.74	74.70 $\pm$ 10.00	74.16 $\pm$ 2.35
(0.8, 0.1, 0.1)	97.32 $\pm$ 0.81	92.42 $\pm$ 1.42	59.91 $\pm$ 10.10
Δ ESG-prio vs. Uniform			
value	-2.85†	-15.60†	+5.59
CI	[-4.20, -1.50]	[-24.39, -8.18]	[-0.64, +11.28]
Δ Sent-prio vs. Uniform			
value	-0.04	+2.12†	-8.66
CI	[-0.43, +0.65]	[+1.62, +2.71]	[-19.77, +0.87]

Priority control exhibits a notable asymmetry. Deprioritizing low-priority classes reliably degrades their performance: ESG-

prio drops Sentiment by -2.85 pp and FLS by -15.60 pp, with both CIs excluding zero, confirming that the priority mechanism is functional and causally connected to allocation. By contrast, increasing priority for high-importance classes yields only marginal gains that are not statistically significant: the ESG-prio lift on ESG is $+5.59$ pp with CI $[-0.64, +11.28]$ just crossing zero (the single-seed $+10.86$ pp figure of an earlier report lies on the upper tail of the resampled distribution). This asymmetry suggests that the baseline already operates near the performance ceiling for high-priority classes, and that priority-aware SemCom may be more effective as a protection mechanism for de-prioritized tasks than as a boosting mechanism for prioritized ones.

Dirichlet-sampled training behaves as a regularizer rather than a runtime lever. A B3-cond variant takes $(\gamma, \mathbf{w})$ as input and trains under stochastic sampling $\mathbf{w} \sim \text{Dir}(\alpha)$ per batch (§III-D), achieving $\text{ESG} \approx 77\%$ averaged across the simplex, which is $+5.5$ pp above the per-priority retrained B3 at uniform $\mathbf{w}$. However, its inference-time Pareto range is narrow: sliding w_{ESG} from 0.1 to 0.9 moves ESG by only about 2.4 pp, versus a 14.25 pp envelope under per-priority retraining. The two interfaces are complementary: Dirichlet training when one robust deployable model is needed, and retraining when the operator wants to actively penalize a de-prioritized task at the cost of T training runs.

C. Adapter vs. Gate at Matched Parameter Budget

The SoftMaskGate is one instantiation of TP-SRA’s per-task readout. To test whether feature-mixing rather than feature-scaling is a stronger inductive bias for priority-aware allocation, we evaluate per-task Houlsby-style adapters (B4) with bottleneck rank $r = 32$ at matched parameter budget (approximately 150K) on AWGN. Mean accuracies across the 10-point SNR sweep are reported in Table IV.

The adapter significantly outperforms the gate on the hardest task: B4 beats B3 on ESG by $+5.29$ pp (CI $[+1.17, +10.07]$, excludes 0), with a small but significant Sentiment loss of -0.75 pp. At matched parameter budget, methods that mix features (Houlsby adapters) outperform methods that scale features (multiplicative gates), suggesting that priority-aware allocation benefits from representational transformation rather than simple reweighting. SNR-conditioned and priority-conditional adapter variants are left to future work.

TABLE IV

ADAPTER VS. GATE, AWGN. MEAN $\pm$ STD ACROSS SNR SWEEP OVER $n = 3$ RANDOM SEEDS. BOTH READOUTS ARE MATCHED AT APPROXIMATELY 150K ADDED PARAMETERS. $\dagger$ MARKS THE ESG DELTA VS. B3 WHOSE PAIRED-BOOTSTRAP 95% CI EXCLUDES 0.

Readout	Sent.	FLS	ESG
B3 SoftMaskGate (proposed)	97.39 $\pm$ 0.86	92.46 $\pm$ 1.76	71.41 $\pm$ 3.23
B4 Housby Adapter	96.64 $\pm$ 0.51	91.00 $\pm$ 2.53	76.70 $\pm$ 1.27 $\dagger$

D. Joint Rate-Mask Results

To test whether exploiting both axes of Theorem 2 closes the cross-channel gap, we train B6: the SoftMaskGate of §III-D paired with a ChannelRateGate that outputs $\mathbf{c} \in [0, 1]^N$ via sigmoid, trained under the L1 rate penalty $\lambda_c = 0.01$ (§III-C). The L1 norm $\|\mathbf{c}\|_1$ decreases monotonically from 31.9/32 at -15 dB to approximately 28/32 at $+20$ dB, confirming a nonzero physical-axis gradient. However, multi-seed evaluation ($n = 3$) shows no ESG improvement on any channel: B6 loses to B1 by $-8.09 / -5.90 / -5.47$ pp on AWGN/Rayleigh/Rician (all CIs exclude 0; Table II for B6 vs. Baseline). Per-symbol pruning patterns are seed-random (mean cross-seed Spearman correlation approximately zero), indicating that the gate learns how much to prune but not what to prune. The simplest joint solver does not close the gap.

E. Statistical Reliability and Limitations

Paired-bootstrap CIs (AWGN, $n=3$) computed by resampling 10,000 (seed, SNR) pairs against Baseline yield the honest reading of Table II: B1’s ESG win is the single statistically robust cross-channel improvement, while B2 and B3 ESG gains are not significant on any channel. On AWGN, expanding the analysis to all three tasks shows that B1 and B2 also significantly degrade FLS (-1.54 pp CI $[-2.71, -0.30]$ and -1.61 pp CI $[-2.93, -0.52]$ respectively), while B3’s only significant gain is on Sentiment ($+0.67$ pp CI $[+0.17, +1.19]$).

We acknowledge two limitations. First, convexity (A3) can be violated by near-linear loss curves on very small datasets, narrowing Theorem 1’s guarantee there. The neural solver does not rely on (A3), and Theorem 2 inherits this caveat with smoothness (A4). Second, we compare against a within-paradigm baseline (B0, [3]), controlling architecture, channel model, and task formulation within a fixed framework. Other forward-looking items are discussed in Section VI.

F. Rate Scaling: The Working Axis

Theorem 2 predicts that semantic and physical axes are coupled. We test this on AWGN by retraining Baseline and B3 at $N=64$ ($n=3$ seeds each) against the corresponding $N=32$ runs (paired-bootstrap CIs, 10,000 resamples). Three findings emerge. First, rate alone is not free: Baseline ESG drops by -3.92 pp from $N=32$ to $N=64$ (CI $[-5.67, -2.13]$). Second, the gate absorbs the rate increase: B3 ESG holds flat at -0.20 pp (CI $[-2.77, +2.31]$). Third, the gate’s value grows with rate: B3 vs. Baseline at $N=64$ is $+5.19$ pp (CI $[+3.64, +6.66]$), substantially larger than $+1.47$ pp at $N=32$. The allocation mechanism’s value is therefore rate-dependent.

G. Bridging the Theoretical Bound and Empirical Results

None of the soft-mask methods reaches the water-filling bound of Theorems 1 and 2. At matched parameter budget, Housby adapters significantly outperform the gate on ESG (Table IV), which suggests that priority-aware allocation requires representational transformation rather than reweighting: a multiplicative gate over an entangled representation cannot express the optimal per-task mapping that Theorem 2’s semantic axis presumes. Combined with §V-F, closing the gap requires both axis-aligned encoders and capacity headroom, neither provided by the simplest joint solver (B6, §V-D).

VI. FUTURE WORK

The most direct extension of TP-SRA is to push allocation upstream of the channel encoder by prepending K learnable task tokens per task to the input sequence. Adapting prefix tuning to the semantic channel, these tokens participate in self-attention within the FinBERT encoder, absorb task-relevant information from the content tokens, traverse the channel alongside them, and are read out by task-specific heads at the decoder. The priority vector $\mathbf{w}$ then controls how many tokens each task receives under a fixed budget K : for instance, $\mathbf{w} = (0.5, 0.3, 0.2)$ with $K = 6$ allocates three tokens to Sentiment, two to FLS, and one to ESG, turning the continuous mask allocation of Theorem 1 into a discrete, interpretable token budget. This directly addresses the entanglement obstacle of §V-G: each task token learns during encoding to attend to the features it needs, so task-relevant information is already separated into dedicated symbols by the time the signal hits the channel. The gate problem dissolves, since the decoder is no longer asked to un-mix a representation that was mixed without task awareness; it was mixed task-specifically from the start.

VII. CONCLUSION

TP-SRA, anchored by Theorems 1 and 2, shows that hard partitioning is the only allocation mechanism delivering statistically significant cross-channel gains under multi-seed evaluation, primarily through variance reduction. Soft and joint-rate learned allocation do not close the gap to the theoretical bound. The design space narrows the architectural requirements: feature mixing outperforms feature scaling at matched parameter budget, while runtime priority interfaces collapse into multi-task regularizers rather than steerable controls.

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